

Statistics & Sampling



Sampling methods

- 1 A school wants to survey students about the cafeteria. Explain why surveying only students in the chess club would produce biased results, and suggest a better sampling method.
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Comparing distributions using shape, center, and spread

- 2 Two data sets have the same median but Data Set A has a much larger range than Data Set B. Explain what this tells you about the variability of each data set.
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Linear regression and correlation

- 3 A scatterplot of hours studied vs. exam score has a line of best fit $y = 6.5x + 52$. Interpret the slope and y -intercept in context.
 - 4 A data set has a correlation coefficient of $r = -0.85$. Describe the strength and direction of the relationship.
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Using a line of best fit to make predictions

- 5 A line of best fit for predicting weight (in kg) from height (in cm) is $W = 0.9H - 70$. Predict the weight of a person who is 175 cm tall, and explain a limitation of using this model for a person who is 250 cm tall.
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Interpreting the standard deviation

- 6 Data Set A has standard deviation 2, and Data Set B has standard deviation 8, both with the same mean. Explain what this indicates about the two data sets, and give a real-world example of each.
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Margin of error and confidence in survey results

- 7 A survey estimates that 42% of a population favors a new policy, with a margin of error of 3%. A second survey with a larger sample size estimates the same statistic with a margin of error of 1.5%. Explain why the second survey's margin of error is smaller, and what this means for the precision of the estimate.
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Evaluating experimental design

- 8 A researcher wants to test whether a new fertilizer increases crop yield. Explain why using a control group (plants without the fertilizer) is essential to this experiment's design.
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Calculating and comparing means from grouped data

- 9 A class of 25 students has a mean test score of 78. A second class of 15 students has a mean score of 88. Find the combined mean score of all 40 students.
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Recognizing bias in survey questions and methods

- 10 A survey asks, "Don't you agree that the excellent new park should be funded?" Explain why this question is biased, and rewrite it in a neutral form.
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Comparing two data sets using boxplot-style summaries

- 11 Data Set A has minimum 10, $Q1 = 20$, median 30, $Q3 = 40$, maximum 50. Data Set B has minimum 15, $Q1 = 28$, median 30, $Q3 = 32$, maximum 45. Both sets have the same median. Compare their spread using the interquartile range (IQR).
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Extrapolation vs. interpolation

- 12 A line of best fit is built from data on shoe size (ranging from 6 to 12) and height. Explain the difference between using this model to predict height for a shoe size of 9 versus a shoe size of 20, referencing interpolation and extrapolation.
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Relative frequency and probability from a frequency table

- 13 A frequency table of exam grades shows: A: 8 students, B: 15 students, C: 12 students, D: 5 students. Find the relative frequency (as a percent) of students who earned a B or higher.
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Random assignment vs. random sampling

- 14 Explain the difference between random sampling and random assignment, and state which one allows a study to make causal claims.

A multi-part regression and residual analysis problem

- 15** This question has three parts. A line of best fit for predicting sales revenue (in thousands of dollars) from advertising spend (in thousands of dollars) is $R = 4.2A + 18$. (a) Predict the revenue for an advertising spend of 10 thousand dollars. (b) If the actual observed revenue at $A = 10$ was 55 thousand dollars, find the residual (actual minus predicted). (c) Explain what a positive residual indicates about this particular data point relative to the line of best fit.